

Notes: Cuddy and Currie (JPE, 2026)

Rules versus Discretion: Treatment of Mental Illness in US Adolescents

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1 Big picture

- **Question.** When adolescents first present with anxiety, depression, or adjustment disorders with anxiety/depression features, how do different initial treatment choices affect later outcomes? In particular, is treatment that follows strict FDA approval better than treatment that follows broader professional guidelines, and how harmful is treatment that follows neither?
- **Setting.** The paper studies privately insured U.S. children and adolescents using a very large national claims database from Blue Cross Blue Shield (BCBS) for 2012–2018, supplemented with IQVIA prescription data. The focus is on *initial* treatment episodes for children ages 10–17.
- **Core challenge.** Treatment is very unlikely to be random. Sicker children, children with different diagnoses, and children living in places with different provider cultures may receive systematically different care. So simple comparisons across treatment groups are not causal.
- **Main empirical design.** The paper uses area-level measures of provider supply and practice style as instruments for treatment, and combines them with a post-Lasso 2SLS approach to deal with the very large instrument set.
- **Main empirical result.** Red-flag prescribing—drugs that are neither FDA approved nor consistent with professional guidelines—leads to worse outcomes: higher total costs, higher ER/hospital use, and more self-harm.
- **Main comparison across guideline types.** FDA-approved drugs appear best at reducing self-harm in the short and medium term, but they are relatively costly. Gray-area prescribing—not FDA approved but consistent with professional guidelines—reduces costs and facility use and does not raise self-harm.
- **Economic and policy takeaway.** Medical guidelines appear to help scale evidence into real-world practice. The strict-vs-flexible guideline question is not trivial: strict FDA rules seem better for self-harm, while broader professional guidelines seem better for costs and facility use. What looks clearly bad is complete discretion in the form of red-flag prescribing.

2 Data and measurement (key objects)

2.1 Observed data

- **Main claims source.** The paper uses the BCBS Axis database, the Blue Cross Blue Shield Alliance for Health Research claims dataset, covering inpatient, outpatient, and pharmaceutical claims for 2012–2018.
- **Supplementary prescription source.** The paper adds IQVIA Longitudinal Prescription Data (LRx), which helps measure prescribing patterns for all prescribers in an area, not only those who see BCBS children.
- **Why claims data are useful here.**
 - They are large and national.
 - They allow children to be followed over time.

- They contain diagnoses and treatments together, so the researcher can classify whether treatment is consistent with evidence-based guidelines.
- Because the children are insured, the paper can largely abstract from uninsured access barriers and focus instead on variation in provider behavior.
- **Initial sample construction.**
 - Start with children observed before age 11 and for at least 1 year between ages 10 and 17.
 - Require a master member ID, pharmacy coverage, valid geography, and consistent demographics.
 - From 2,201,566 BCBS-covered children meeting these conditions, 202,066 have at least one mental-health-related claim.
- **Final analysis sample.** The final sample is 45,223 children who can be followed for at least 24 months and whose initial diagnoses are depression, anxiety, or adjustment disorders with anxiety/depression features.
- **Follow-up windows.** Outcomes are measured 3 months, 12 months, and 24 months after the initial claim. Longer follow-up is possible for many children, but requiring longer horizons would sharply reduce sample size.

2.2 Treatment categories and key definitions

- **Initial episode.** The paper focuses on the first mental health episode because guidelines are relatively clear at the start of treatment, and because early treatment can affect later trajectories.
- **Broad treatment margin.**
 - **No treatment** in the 3 months after the initial claim.
 - **Any treatment** in the 3 months after the initial claim.
- **Conditional on treatment.**
 - **Therapy only**
 - **Any drug treatment**
- **Conditional on drug treatment.**
 - **FDA-approved drugs:** drugs approved by the FDA for that disorder/age group.
 - **Gray-area drugs:** not FDA approved for that use, but consistent with professional guidelines.
 - **Red-flag drugs:** neither FDA approved nor consistent with professional guidelines.
- **Examples of red-flag drugs.** The paper highlights antipsychotics, tricyclic and other non-SSRI antidepressants, and long-term benzodiazepines as the most common red-flag classes.
- **Interpretation of the three categories.**
 - FDA-approved treatment corresponds to a *strict* evidence rule.
 - Gray-area treatment corresponds to *broader professional discretion* constrained by professional guidance.

- Red-flag treatment corresponds to *complete discretion* outside both FDA and professional rules.

2.3 Key descriptive objects

- **Treatment shares in the final sample.**
 - 75% receive some treatment within 3 months.
 - 57% receive therapy only.
 - 28% receive some drug treatment.
 - 10% receive gray-area drugs.
 - 4% receive FDA-approved drugs.
 - 4% receive red-flag drugs.
- **Sample composition.**
 - 56.7% of the analysis sample are female.
 - Mean age at first mental illness episode is about 11.62.
 - 36.7% have a neurodevelopmental condition, and 26.4% have ADHD.
- **Initial severity indicators.**
 - About 1.6% are hospitalized during the first mental illness episode.
 - About 3.5% have an ER visit during the first episode.
 - 54.4% have an evaluation visit at the first episode.
- **Diagnosis prevalence and overlap.** Anxiety is the most common diagnosis in the sample. Figure 1 also shows substantial overlap between anxiety, depression, and adjustment disorders, which is one reason why the paper treats diagnosis carefully in the controls.
- **Raw outcome patterns.** Children who receive drugs appear, in raw means, to have worse outcomes than children receiving therapy only. The authors explicitly warn that these are not causal patterns, because children receiving drug treatment may be more severely ill.

2.4 Outcomes, markets, and timing

- **Main outcomes.**
 - log total health care costs,
 - an indicator for facility use (ER visit or hospitalization),
 - an indicator for any self-harm behavior.
- **Supplementary outcomes.**
 - log total mental health care costs,
 - length of facility stay.
- **Why these outcomes matter.**

- Total costs capture not just mental health care but also physical consequences of treatment, such as injuries or adverse drug reactions.
- Facility use is important because mental health crises often result in ER visits or hospitalization.
- Self-harm is central because antidepressants and related psychiatric medications are closely monitored for suicidality risk in children.
- **Market definition.** A child’s relevant local mental health market is defined using actual travel patterns: for each ZCTA, the authors use up to the 10 most common ZCTAs where local children receive mental health care. This creates treatment markets based on real provider access rather than arbitrary county boundaries.
- **Leave-one-out construction.** The provider-supply measures are constructed leaving out the index child, which helps avoid mechanical reflection problems.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline outcome equation

The paper begins with baseline linear probability / OLS models of the form

$$\begin{aligned} Outcome_{ymzi} = & \beta_0 + \beta_1 Treatment_{ymzi} + \beta_2 Female_{ymzi} + \Omega \cdot Age_{ymzi} \\ & + \beta_3 Hosp0_{ymzi} + \beta_4 ER0_{ymzi} + \beta_5 Eval0_{ymzi} + \beta_6 HospPast6m_{ymzi} \\ & + \alpha \cdot Diagnosis_{ymzi} + \beta_7 Neurodev_{ymzi} + \mu_m + \gamma_y + \delta_z + \varepsilon_{ymzi}. \end{aligned} \quad (1)$$

Here:

- $Outcome_{ymzi}$ is one of the main outcomes,
- $Treatment_{ymzi}$ is an indicator for a particular treatment margin,
- Age is a vector of single-year age fixed effects,
- $Hosp0$, $ER0$, and $Eval0$ capture initial severity / episode type,
- $HospPast6m$ captures recent hospital use,
- $Diagnosis$ includes depression, anxiety, and adjustment-disorder indicators,
- $Neurodev$ indicates a neurodevelopmental condition,
- μ_m , γ_y , and δ_z are month, year, and area fixed effects.

Interpretation. This is the paper’s baseline selection-on-observables specification. It tries to compare children with similar demographics, diagnoses, severity, and area/time environment. But the paper is explicit that this is *not enough* for causal identification because treatment likely still correlates with unobserved severity and preferences.

3.2 Nested treatment comparisons

A useful way to read the empirical design is as a sequence of increasingly fine decisions:

1. **Any treatment vs. no treatment**
2. **Drug treatment vs. therapy only**, conditional on treatment
3. **Red-flag vs. other drugs**, conditional on drug treatment
4. **Gray-area vs. other drugs**, conditional on drug treatment
5. **FDA-approved vs. other drugs**, conditional on drug treatment

So the paper is not estimating one huge multinomial system. Instead, it studies the relevant margins step by step. This is why the figures later can be drawn as treatment decision trees.

3.3 Instrumental variables: area-level provider supply and practice style

The paper’s main identification idea is that children in different places are exposed to systematically different local treatment cultures.

The eight underlying small-area variables are:

1. share of BCBS mental health providers who are psychiatrists,
2. share of BCBS mental health providers who are PCPs,
3. among PCPs, share of each physician’s child caseload receiving psychiatric drugs,
4. among all IQVIA first psychiatric prescriptions, share written by psychiatrists,
5. share of BCBS anxiety/depression patients prescribed antipsychotics,
6. psychiatrists per 1,000 BCBS children ages 10–17,
7. therapists per 1,000 BCBS children ages 10–17,
8. PCPs providing mental health treatment per 1,000 BCBS children ages 10–17.

Interpretation. Some of these are *provider supply* measures and others are *practice style* measures. The key idea is that children do not choose these local prescribing cultures. So, conditional on observables and area fixed effects, the local style can serve as an exogenous shifter of treatment.

3.4 Why area-level rather than physician-level instruments?

The paper emphasizes that physician-level propensities to prescribe can be problematic because patient-physician matching is endogenous. If, for example, highly informed or highly motivated parents seek out certain physicians, then a doctor’s own prescribing tendency is not cleanly exogenous.

Area-level practice style is meant to be more plausibly exogenous:

$$Treatment_{ymzi} = \Pi(Z_z \times X_i) + X_i\Gamma + FE + u_{ymzi},$$

where Z_z denotes local supply/practice-style measures and X_i denotes child characteristics.

Interpretation. Children living in places with particular provider mixes and prescribing styles are more likely to receive some kinds of care. The identifying variation comes from place-level differences in treatment environment, not from the child’s chosen physician.

3.5 Post-Lasso 2SLS

Because the instrument set is high-dimensional, the paper uses post-Lasso 2SLS.

- The full potential instrument vector contains a second-degree polynomial in the core practice-style measures and interactions with age, sex, severity, and diagnosis.
- After excluding provider-count measures that Lasso never chooses, the potential vector still contains 116 candidate instruments.
- Lasso is forced to keep the main control set in the first stage.
- The selected instruments are then used in an OLS first stage, and the predicted treatment is plugged into the second-stage outcome equation.

Formally, the second stage is

$$Outcome_{ymzi} = \beta_0 + \beta_1 \widehat{Treatment}_{ymzi} + X'_{ymzi} \theta + \mu_m + \gamma_y + \delta_z + \varepsilon_{ymzi}. \quad (2)$$

Interpretation. The key purpose of post-Lasso is not just variable selection for convenience. It is to reduce weak-instrument problems in a setting where the raw instrument space is too large.

3.6 First-stage fit and interpretation

The paper reports Montiel Olea–Pflueger effective F -statistics for the preferred first stages:

$$FEFF \in [14.93, 30.28].$$

These are reported for the different treatment margins and indicate that, once post-Lasso chooses the relevant subset, the first stages pass weak-instrument tests.

Interpretation. The paper is very careful here. The raw supply variables by themselves are not strong enough. The identifying content comes from using the right subset of area-style measures and their interactions with child characteristics.

3.7 What the IV coefficients mean

The post-Lasso 2SLS estimates should be read as local average treatment effects for children whose initial treatment is shifted by local practice style and provider mix.

This matters in two ways:

- The IV estimates can be larger than OLS because the affected children are the *marginal* children whose treatment changes when practice style changes.
- The “treatment” coefficients can partly reflect improved detection or reporting, especially for self-harm and facility use.

Interpretation. The paper is not claiming that every treated child becomes worse off along every margin. Rather, it is identifying what happens for the children whose treatment choice is moved by local supply-side variation.

3.8 Supplementary specifications

The paper also studies:

- mental health costs instead of total costs,
- facility length of stay,
- a trimmed sample excluding psychosis, bipolar disorder, autism spectrum disorder, and Tourette syndrome,
- alternative instrument sets, including top-3 selected instruments and a broader set reintroducing provider-count measures.

Interpretation. These are used to check whether the main message—especially the harmfulness of red-flag prescribing and the relative attractiveness of gray-area treatment—is robust to reasonable alternative samples and first-stage choices.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the standard comparison fails

- More severely ill children are more likely to receive drug treatment.
- Conditional on drugs, children with more complicated psychiatric profiles may be more likely to receive red-flag drugs.
- Parents can differ in aggressiveness or urgency of treatment demand.
- Doctors differ in practice style and diagnostic skill.

Any one of these channels can bias OLS comparisons.

Bottom line. A raw comparison of therapy versus drugs, or FDA versus red-flag treatment, is not causal.

4.2 What identifies the treatment effects

The identifying source of variation is that children in different local treatment markets face different provider mixes and prescribing styles:

- more psychiatrists versus more PCP-heavy mental health care,
- more psychiatrist-written first prescriptions,
- more antipsychotic-heavy local practice norms,

- more or fewer local behavioral health professionals.

The paper assumes that these variables shift treatment patterns but, conditional on the rich control set and area fixed effects, are not directly related to the child’s future outcomes except through treatment.

Interpretation. The paper is using local “rules of thumb” in medicine as quasi-random variation in actual care.

4.3 Why first episodes help credibility

The focus on the first 3 months after the first mental health claim makes the strategy more credible.

- Treatment guidelines are clearest at the start of care.
- Children and parents have less accumulated treatment history, so there is less scope for prior individualized experimentation to drive treatment.
- Parents are less likely to have precise drug preferences in an initial episode.

Interpretation. Initial treatment is the place where area-level practice style is most likely to matter, and where “rules versus discretion” is most directly testable.

4.4 Why the market definition is thoughtful

The paper does not define treatment markets mechanically by county or hospital referral region. Instead, it constructs small areas using actual travel flows of BCBS children to mental health providers, including up to 10 ZCTAs per market.

Why this helps.

- It includes the providers that are actually relevant to children in that zip code.
- It scales naturally in urban vs. rural areas.
- It avoids arbitrary assignment of providers to one market only.
- It is specific to psychiatric care rather than health care in general.

4.5 The Lasso step: why it matters

The paper’s instrument strategy would be much weaker without Lasso. The authors explicitly show that:

- the raw supply-side instruments alone are not strong enough,
- the full interacted instrument vector is too large,
- but the *selected* subset passes weak-instrument tests.

Interpretation. This paper is a good example of machine-learning-assisted IV: not using Lasso for prediction per se, but to recover a strong and disciplined first stage from many potential shifters.

4.6 Key limitations

- The sample covers privately insured children, not the uninsured or Medicaid populations.
- The strategy identifies local average treatment effects, not universal average effects.
- Practice-style instruments could still be questioned if unobserved local shocks to outcomes are correlated with treatment style.
- Claims data observe treatment and subsequent utilization, but not all underlying clinical detail.
- Some estimated increases in self-harm or facility use may reflect increased detection rather than pure worsening of underlying mental health.

Interpretation. These limitations matter, but they do not obviously overturn the paper’s core ranking: red-flag prescribing looks worst, while FDA-approved and gray-area treatments both dominate it on key margins.

5 Tables and figures

5.1 Figure 1 and Table 1: who is in the sample and how treatment is categorized

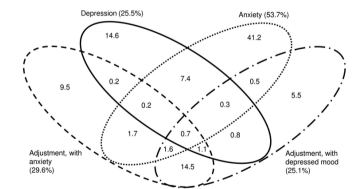


FIG. 1.—Prevalence of depression, anxiety, and adjustment disorders among sample. The figure shows the prevalence of depression (solid line), anxiety (dotted line), and adjustment disorders with features of depression (dash-dotted line) or anxiety (dashed line) among the 45,223 children in the sample. Intersecting ellipses correspond to sample children who have those corresponding diagnoses, where the number listed in each area represents the percentage of the total sample with that combination of diagnoses. Each diagnosis is defined according to the DSM-5.

TABLE 1 DISCRETE TREATMENTS									
	All Mental Health BCBS	All Mental Health BCBS	Sample	Therapy Only	Therapy and Drugs	Drugs Only	FDA-Approved Drugs	Gray Area Drugs	Red-Flag Drugs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
A. Child Characteristics									
Number of children	2,281,566	302,666	45,223	58,312	55,649	4,481	9,108	1,794	1,994
% of sample	...	...	...	...	...	...	...	...	...
Female	486	128	167	175	171	115	188	4	58
% of the population in sample	1,835	8,326	8,728	4,752	4,681	4,382	4,671	4,386	4,654
Neurodevelopmental condition	175	378	307	375	345	487	464	417	402
Neurodevelopmental condition in 2010	120	271	261	260	266	361	342	301	277
Age at first mental health episode	15,000	11,018	11,019	11,512	11,977	11,583	12,311	11,770	12,058
Time since first mental health episode	...	...	...	...	...	...	...	...	...
First mental health episode in 2010	...	...	...	...	...	...	...	...	...
First mental health episode in 2011	...	...	...	...	...	...	...	...	...
B. Child Outcomes									
Depressed, any reason	385	385	405	394	392	388	388	395	312
Anxious	387	387	405	312	302	388	388	315	305
AD, any reason	389	314	418	312	317	388	387	326	321
Average monthly costs (2011-12)	127	302	322	331	378	306	404	420	461

C. Prevalence of Lifetime Depression/Anxiety Mental Health Diagnoses in Sample									
Depression	...	...	...	...	...	...	...	...	...
Anxiety	...	...	...	...	...	...	...	...	...
Adjustment with anxiety	...	...	...	...	...	...	...	...	...
Adjustment with depression	...	...	...	...	...	...	...	...	...
Adjustment with anxiety/Depression	...	...	...	...	...	...	...	...	...

Notes: Data are from the BCBS data database of insurance claims for 2010-12. It covers children who have a valid master number ID, pharmacy coverage, and valid geographic information and who were observed both before age 11 and for at least 1 year between ages 10 and 18. Column 1 includes children who had at least one claim related to mental illness. Column 2 includes all children who can be followed for at least 36 months and have a diagnosis of depression, anxiety, or adjustment disorders with features of anxiety and depression, and. It includes all children who can be followed for at least 36 months. The size population of children in the sample. Column 3 includes those who received any follow-up treatment in the 3 months following an initial claim. Column 4 includes children who received only therapy (no drugs) in the 3 months following the initial claim. Column 5 includes children who received both therapy and drugs in the 3 months following the initial claim. Column 6 includes children who received only drug treatment (no therapy) in the 3 months following the initial claim. Column 7 includes children who received only FDA-approved drugs in the 3 months following the first claim. Column 8 includes children who received gray-area drugs in the 3 months following the first claim. Column 9 includes children who received red-flag drugs in the 3 months following the first claim. Columns 4-10 are mutually exclusive categories.

Read:

- Figure 1 shows that anxiety, depression, and adjustment disorders overlap a lot. So this is not a clean single-diagnosis world; comorbidity is common.
- Table 1 shows the basic treatment split: 75% are treated, 57% get therapy only, and 28% get some drug treatment.
- The striking descriptive fact is that gray-area prescribing is much more common than FDA-approved prescribing: 10% of the sample receives gray-area drugs, versus only 4% FDA-approved drugs.
- Red-flag prescribing is not rare: it is about as common as FDA-approved prescribing in the sample.
- This is already suggestive of the paper’s central message: in practice, professional guidelines matter more than FDA approval for what doctors actually do.

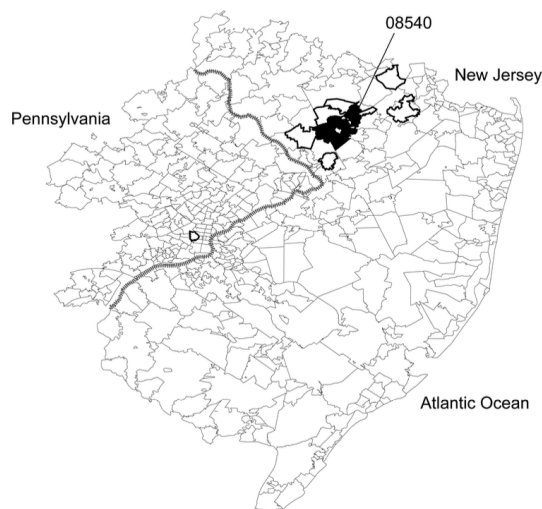


FIG. 2.—Definition of market area for specific zip code: 08540 (Princeton, NJ). Children residing in Princeton saw providers for mental health services in their own zip code (08540; shaded area) and also in zip codes outlined by black lines. Hence, these zip codes together are considered the relevant market area for children in Princeton. Markets for all other zip codes are defined analogously so that providers in each zip code may form part of the market for children residing in multiple zip codes. We include up to 10 ZCTAs in each market ranked in terms of the number of children who travel there (as in this example); in markets where children traveled to fewer than 10 ZCTAs, we use only those ZCTAs.

TABLE 2
SMALL-AREA VARIATION IN TREATMENT, PROVIDER SUPPLY, AND PRACTICE STYLE

	Percentile of Area-Level Distribution					Within-ZCTA Variation (%)	Between-ZCTA Variation (%)
	10th (1)	25th (2)	50th (3)	75th (4)	90th (5)	(6)	(7)
A. Treatment (Source Is BCBS)							
Child treated within 3 months	.50	.65	.75	.83	.89	79	21
If any treatment:							
Therapy only	.50	.65	.76	.85	.93	63	37
Drugs and therapy	.00	.06	.12	.17	.25	73	27
Drugs only	.00	.03	.10	.19	.31	67	33
Drugs	.05	.15	.24	.35	.50	65	35
If any drug treatment:							
Red-flag drug	.00	.00	.20	.33	.50	79	21
Gray area drug	.00	.33	.50	.68	1.00	76	24
FDA-approved drug	.00	.00	.14	.29	.50	77	23
B. Provider Supply (Source Is BCBS)							
Providers per 1,000 BCBS children ages 10–17:							
Psychiatrists	4.66	6.19	8.85	12.17	16.93	16	84
Therapists	16.26	21.87	29.33	41.45	54.67	13	87
PCPs providing mental health treatment	24.42	34.06	47.38	63.07	87.07	28	72
C. Practice Style (Source Is BCBS or IQVIA)							
BCBS:							
Share of PCP patients who receive mental health drug	.04	.05	.07	.09	.12	21	79
Share of mental health providers who are PCPs	.13	.15	.18	.21	.23	12	88
Share of mental health providers who are psychiatrists	.08	.10	.12	.14	.17	14	86
Share of depression and anxiety patients with antipsychotics	.06	.07	.08	.09	.11	51	49
IQVIA:							
Share of new prescriptions by psychiatrists	.03	.05	.07	.09	.13	33	67

NOTE.—The table is calculated by computing small area-level rates and then calculating percentiles of the distributions of those rates. Small areas are defined using information about where children in a particular zip code actually go to receive mental health care. Each row represents a separate distribution of ZCTAs. For example, places at the 90th percentile in terms of psychiatrists per capita could be at the 10th percentile in terms of the fraction of PCPs treating mental health. BCBS indicates that the variable is calculated using our main sample. IQVIA indicates that the data were calculated using that database. The variables within- and between-ZCTA variation represent the decomposition of total variation in each treatment, supply, and style measure.

5.2 Figure 2 and Table 2: market definition and local variation

Read:

- Figure 2 gives the Princeton, New Jersey example and makes clear that the paper’s “market” is based on where children actually go for care.
- Table 2 shows there is very large geographic variation in treatment and prescribing style.
- The fraction treated within 3 months ranges from 50% at the 10th percentile ZCTA to 89% at the 90th percentile.
- Among drug-treated children, the share receiving red-flag drugs ranges from 0 to 50% across local markets.
- Importantly, much of the variation is within-ZCTA rather than purely across-ZCTA, especially for red-flag treatment. That is exactly what the instrument strategy needs once area fixed effects are included.

5.3 Table 3: first-stage evidence

TABLE 3
FIRST STAGE WITH LASSO SELECTED INSTRUMENTS

	Treated (1)	Drugs (2)	Red-Flag (3)	Gray Area (4)	FDA Approved (5)
Child female	.018*** (.004)	.007** (.004)	.003 (.011)	.004 (.011)	-.007 (.009)
Age 11	.010* (.005)	.020*** (.006)	-.013 (.014)	.024 (.016)	-.012 (.012)
Age 12	.003 (.006)	.047*** (.007)	-.007 (.013)	-.044** (.018)	.051*** (.016)
Age 13	.017** (.007)	.077*** (.007)	.014 (.015)	-.062*** (.018)	.045*** (.014)
Age 14	.021*** (.008)	.110*** (.009)	.022 (.019)	-.069*** (.021)	.044** (.019)
Age 15	.041*** (.013)	.162*** (.013)	.041 (.027)	-.076** (.032)	.034 (.026)
First claim is hospitalization	.034** (.017)	.203*** (.021)	.138*** (.029)	-.095*** (.027)	-.039 (.026)
First claim is ER visit	-.066*** (.023)	-.015 (.015)	.023 (.026)	-.037 (.030)	.015 (.026)
First claim is mental health evaluation	.197*** (.006)	-.117*** (.021)	-.017 (.011)	.022* (.011)	-.004 (.010)
Hospitalized past 6 months	-.019 (.029)	.093*** (.030)	.225*** (.054)	-.208*** (.056)	-.019 (.043)
Depression	.109*** (.007)	.200*** (.008)	-.001 (.020)	-.083** (.033)	.127*** (.017)
Anxiety	.108*** (.006)	.080*** (.020)	-.041** (.016)	.115*** (.021)	-.045*** (.014)
Adjustment with anxiety	.109*** (.009)	.007 (.008)	-.045** (.023)	.055* (.032)	-.005 (.018)
Adjustment with depression	.107*** (.009)	.010 (.010)	.018 (.025)	-.001 (.028)	-.015 (.024)
Adjustment with depression/ anxiety	.134*** (.008)	-.001 (.008)	-.082*** (.019)	.017 (.023)	.069*** (.022)
Neurodevelopmental	.014*** (.005)	.054*** (.005)	.061*** (.010)	-.014 (.012)	-.048*** (.009)
Share of new prescriptions by psychiatrists × Multiple diagnoses	.328*** (.092)				
Only anxiety diagnosis	-.343*** (.119)				
Share of mental health providers who are PCPs × First claim is evaluation	-.583*** (.108)				
Any depression diagnosis				-.178 (.177)	
Only adjustment diagnosis			.590* (.300)	-.681*** (.188)	
Anxiety diagnosis		.721*** (.110)			
Only anxiety diagnosis				.069 (.172)	

TABLE 3 (Continued)

	Treated (1)	Drugs (2)	Red-Flag (3)	Gray Area (4)	FDA Approved (5)
Share of mental health providers who are psychiatrists × any depression diagnosis			.541*** (.170)	-.471*** (.181)	
Share of PCP patients who receive a mental health drug × First claim is evaluation		-.424*** (.121)			
Multiple diagnoses	.276*** (.083)			-.255 (.207)	
Any depression diagnosis					.899*** (.196)
Share of depression and anxiety patients with antipsychotic prescription × Multiple diagnoses	.111 (.085)				
Any depression diagnosis				-.586** (.248)	
Only adjustment diagnosis			1.334* (.695)		
Constant	.330*** (.016)	.295*** (.017)	.196*** (.036)	.655*** (.049)	.110*** (.027)
Observations	45,223	33,812	8,188	8,188	8,188
R ²	.126	.288	.077	.153	.127
F _{EFF}	22.448	30.276	24.928	14.929	20.935
Mean dependent variable	.748	.242	.244	.543	.213

NOTE.—The table presents first-stage results, where a child’s treatment status is instrumented using patient characteristics and the set of instrumental variables selected by the Lasso procedure described by Belloni et al. (2012) using their refined data-driven penalty. The times symbol indicates an interaction between one of the six primary instruments and the child’s diagnosis or severity. Each column is from a single regression model, where the sample is conditional on treatment in col. 2 and conditional on drug treatment in cols. 3–5. F_{EFF} represents the effective F-statistics for the instruments from Montiel Olea and Ploger (2013). All models include area, month, and year fixed effects corresponding to the child’s initial mental illness claim. Standard errors (in parentheses) are clustered at the three-digit zip code level.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

Read:

- Table 3 asks whether the selected instruments actually move treatment choices. They do.
- Girls and older children are more likely to receive treatment and, conditional on treatment, more likely to receive drugs.
- Initial hospitalization strongly predicts later drug treatment and especially red-flag prescribing.
- Children whose first mental-health interaction is only an evaluation are more likely to receive some treatment but less likely to receive drugs.
- Depression increases the probability of FDA-approved treatment; anxiety decreases it.
- The selected first-stage specifications have effective F -statistics roughly between 15 and 30, so the chosen instrument set is not weak.
- The authors also caution against reading too much into any single selected interaction term because Lasso selection can vary in finite samples.

5.4 Table 4: the main causal results

	Log Total Costs			1 (Visited Facility)			1 (Any Harm Behavior)		
	3 Months (1)	1 Year (2)	2 Years (3)	3 Months (4)	1 Year (5)	2 Years (6)	3 Months (7)	1 Year (8)	2 Years (9)
A. Effect of Treatment									
Observations	45,223	45,223	45,223	45,223	45,223	45,223	45,223	45,223	45,223
Mean dependent variable	6.926	7.846	8.448	.112	.182	.254	.025	.096	.052
OLS	.562*** (.016)	.489*** (.016)	.381*** (.016)	.013*** (.003)	.024*** (.004)	.027*** (.005)	.013*** (.001)	.018*** (.002)	.020*** (.002)
R^2	.232	.191	.159	.267	.206	.208	.178	.149	.149
Post-Lasso 2SLS	.584*** (.303)	.552*** (.277)	.1,082*** (.245)	.354*** (.078)	.469*** (.091)	.631*** (.121)	.213*** (.055)	.253*** (.065)	.273*** (.073)
B. Effect of Drug Treatment									
Observations	33,812	33,812	33,812	33,812	33,812	33,812	33,812	33,812	33,812
Mean dependent variable	7.079	7.976	8.550	.113	.184	.258	.029	.042	.058
OLS	.385*** (.020)	.359*** (.018)	.385*** (.017)	.086*** (.005)	.104*** (.007)	.110*** (.007)	.034*** (.004)	.047*** (.004)	.052*** (.004)
R^2	.220	.190	.168	.401	.280	.205	.236	.205	.174
Post-Lasso 2SLS	-.529*** (.218)	-.501*** (.217)	-.487*** (.233)	.021 (.058)	-.041 (.074)	-.115 (.093)	-.042 (.040)	-.005 (.048)	-.096* (.057)

C. Effect of Red-Flag Drug Treatment									
Observations	8,188	8,188	8,188	8,188	8,188	8,188	8,188	8,188	8,188
Mean dependent variable	7.478	8.388	8.945	.226	.315	.394	.085	.112	.136
OLS	.562*** (.035)	.472*** (.035)	.405*** (.035)	.122*** (.010)	.111*** (.012)	.102*** (.012)	.051*** (.007)	.055*** (.006)	.055*** (.009)
R^2	.324	.269	.232	.351	.258	.214	.269	.252	.236
Post-Lasso 2SLS	1.135*** (.310)	1.388*** (.288)	1.458*** (.286)	.217*** (.084)	.266*** (.103)	.457*** (.116)	.186*** (.067)	.189*** (.074)	.149* (.080)
D. Effect of Gray Area Drug Treatment									
Observations	-.310*** (.027)	-.241*** (.026)	-.218*** (.027)	-.074*** (.009)	-.067*** (.010)	-.024*** (.006)	-.024*** (.006)	-.030*** (.007)	-.033*** (.008)
R^2	.307	.233	.220	.343	.232	.211	.264	.248	.234
Post-Lasso 2SLS	-.962*** (.301)	-.1,249*** (.295)	-.1,271*** (.281)	-.170*** (.081)	-.240*** (.101)	-.450*** (.113)	-.089 (.067)	-.097 (.077)	-.061 (.083)
E. Effect of FDA-Approved Drug Treatment									
Observations	-.201*** (.033)	-.205*** (.031)	-.159*** (.031)	-.055*** (.011)	-.054*** (.012)	-.024 (.015)	-.025*** (.008)	-.021** (.009)	-.017 (.011)
R^2	.209	.249	.216	.338	.249	.207	.264	.247	.232
Post-Lasso 2SLS	.288 (.376)	.199* (.096)	1.645** (.646)	-.234 (.175)	-.206 (.195)	.063 (.208)	-.299* (.169)	-.378** (.159)	-.301 (.210)

NOTE.—The table presents results from OLS and post-Lasso 2SLS models of mental health treatment on log total health costs (cols. 1–3), an indicator for facility use (cols. 4–6), and an indicator for any self-harm behavior (cols. 7–9). The instruments in the post-Lasso 2SLS models are chosen using a refined data-driven penalty and are displayed in table A6. Number of instruments = 4 (panel A), 5 (panel B), 5 (panel C), 6 (panel D), 1 (panel E). Postestimation results from the first stage of each model are included in table 3. All models include patient controls (age, female, severity, and diagnosis) as well as area, month, and year fixed effects corresponding to the child's initial mental illness claim. Standard errors (in parentheses) are clustered at the three-digit zip code level.
* $p < .10$.
** $p < .05$.
*** $p < .01$.

Read:

- **Panel A (any treatment).** Post-Lasso 2SLS estimates imply that treatment raises total costs, facility use, and self-harm over 3, 12, and 24 months. The authors interpret this cautiously because treatment may increase detection and because the LATE applies to marginal children.
- **Panel B (drug treatment among the treated).** Once selection into drug treatment is instrumented for, drugs reduce costs relative to therapy alone, with little evidence of worse facility use or self-harm.
- **Panel C (red-flag drugs).** This is the strongest result in the paper. Relative to other drug treatments, red-flag prescribing raises costs, facility use, and self-harm. This is the clearest evidence against unconstrained discretion.
- **Panel D (gray-area drugs).** Gray-area prescribing reduces costs and facility use and has no detectable effect on self-harm. This is the strongest evidence in favor of professional-society flexibility.
- **Panel E (FDA-approved drugs).** FDA-approved prescribing raises costs but lowers self-harm in the short and medium term. This fits the idea that FDA approval especially prioritizes avoiding harmful side effects.

5.5 Figures 3, 4, and 5: treatment trees at the 24-month horizon

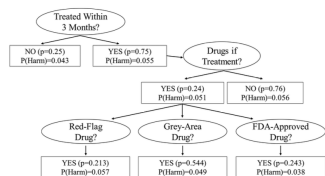


FIG. 3.—Estimated effects of treatment on probability of self-harm at 24 months. The figure traces out the implications of each initial treatment choice on the probability of self-harm at 24 months, where the share of children in each branch (p) is in parentheses. Self-harm estimates are based on the post-Lasso 2SLS estimates shown in table 4.

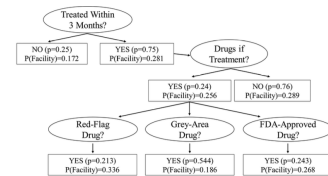


FIG. 4.—Estimated effects of treatment on probability of facility use at 24 months. The figure traces out the implications of each initial treatment choice on the probability of facility use at 24 months, where the share of children in each branch (p) is in parentheses. Facility use estimates are based on the post-Lasso 2SLS estimates shown in table 4.

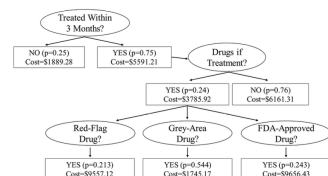


FIG. 5.—Estimated effects of treatment on total cost at 24 months. The figure traces out the implications of each initial treatment choice on log total costs at 24 months, where the share of children in each branch (p) is in parentheses. Log total cost estimates are based on the post-Lasso 2SLS estimates shown in table 4.

Read:

- These figures turn the regression coefficients into intuitive treatment trees.
- **Figure 3 (self-harm).** Among treated children, therapy-only implies about a 5.6% self-harm probability, drug treatment about 5.1%. Within drug treatment, red-flag is highest at 5.7%, gray-area is 4.9%, and FDA-approved is lowest at 3.8%.
- **Figure 4 (facility use).** Treatment raises facility use overall (28.1% vs. 17.2%). Conditional on drugs, red-flag is worst at 33.6%, gray-area is best at 18.6%, and FDA-approved is in between at 26.8%.
- **Figure 5 (total costs).** Treatment raises 24-month total costs overall, but much of this is driven by therapy-only treatment. Conditional on being treated, drug treatment is cheaper than therapy only. Within drugs, gray-area is cheapest, while red-flag and FDA-approved are both much more expensive.

5.6 Table 5: supplementary outcomes

	Log Total Mental Health Costs			Length of Facility Stay		
	3 Months (1)	1 Year (2)	2 Years (3)	3 Months (4)	1 Year (5)	2 Years (6)
A. Effect of Treatment						
Observations	45,223	45,223	45,223	45,223	45,223	45,223
Mean dependent variable	6.082	6.566	6.890	.437	.775	1.240
OLS	.897*** (.022)	1.103*** (.022)	1.100*** (.023)	.327** (.160)	.537*** (.182)	.778*** (.209)
R ²	.308	.268	.266	.006	.033	.028
Post-Lasso 2SLS	1.384*** (.361)	1.585*** (.379)	1.638*** (.370)	.959 (2.475)	2.791 (3.024)	7.253 (4.799)
B. Effect of Drug Treatment						
Observations	33,812	33,812	33,812	33,812	33,812	33,812
Mean dependent variable	6.352	6.881	7.201	.501	.888	1.412
OLS	-.054 (.027)	-.174*** (.025)	-.294*** (.027)	.870*** (.250)	1.272*** (.319)	1.714*** (.405)
R ²	.239	.214	.198	.026	.094	.030
Post-Lasso 2SLS	-1.589*** (.309)	-1.472*** (.349)	-1.390*** (.366)	-.174 (1.120)	-1.572 (1.606)	-6.217* (3.411)

	C. Effect of Red-Flag Drug Treatment					
	Observations	Mean dependent variable	OLS	Post-Lasso 2SLS	OLS	Post-Lasso 2SLS
Observations	8,188	8,188	8,188	8,188	8,188	8,188
Mean dependent variable	6.388	7.098	7.222	1.655	2.581	3.656
OLS	.464*** (.040)	.469*** (.041)	.444*** (.041)	2.361*** (.792)	3.478*** (.725)	4.462*** (.938)
R ²	.425	.376	.338	.062	.081	.084
Post-Lasso 2SLS	1.514*** (.320)	1.553*** (.349)	1.345*** (.365)	1.610 (3.968)	5.649 (4.810)	13.130* (7.152)
D. Effect of Gray Area Drug Treatment						
Observations	8,188	8,188	8,188	8,188	8,188	8,188
Mean dependent variable	6.388	7.098	7.222	1.655	2.581	3.656
OLS	-.164*** (.030)	-.124*** (.037)	-.074*** (.030)	-1.218*** (.365)	-1.681*** (.285)	-2.109*** (.402)
R ²	.415	.366	.328	.061	.078	.081
Post-Lasso 2SLS	-1.984*** (.348)	-2.034*** (.374)	-1.760*** (.380)	-1.115 (3.304)	-4.516 (3.774)	-12.166** (5.874)
E. Effect of FDA-Approved Drug Treatment						
Observations	8,188	8,188	8,188	8,188	8,188	8,188
Mean dependent variable	6.388	7.098	7.222	1.655	2.581	3.656
OLS	-.300*** (.039)	-.360*** (.041)	-.413*** (.042)	-1.006** (.459)	-1.644*** (.583)	-2.179*** (.784)
R ²	.418	.371	.335	.061	.078	.081
Post-Lasso 2SLS	1.267* (.716)	1.244* (.750)	1.278 (.824)	-15.038 (11.842)	-15.917 (13.868)	-2.715 (17.102)

NOTE.—The table presents results from OLS and post-Lasso 2SLS models of mental health treatment on log total mental health costs (cols. 1–3) and length of facility stay (cols. 4–6). The instruments in the post-Lasso 2SLS models are chosen using a refined data-driven penalty and are displayed in table A6. Number of instruments = 4 (panel A), 3 (panel B), 3 (panel C), 6 (panel D), 1 (panel E). Postestimation results from the first stage of each model are included in table 5. All models include patient controls (age, female, severity, and diagnosis) as well as area, month, and year fixed effects corresponding to the child's initial mental illness claim. Standard errors (in parentheses) are clustered at the three-digit zip code level.
* $p < .10$.
** $p < .05$.
*** $p < .01$.

Read:

- Table 5 repeats the analysis for mental-health-specific costs and facility-stay length.
- The pattern for mental health costs reinforces the main results from total costs: drug treatment is cheaper than therapy alone, red-flag is expensive, and gray-area is the cheapest.
- Length of stay estimates are noisy. So the strongest utilization evidence is about whether children reach a facility, not about how long they stay there.

TABLE 6
EFFECTS OF TREATMENT ON COSTS, FACILITY USE, AND SELF-HARM AMONG CHILDREN WITHOUT PSYCHOSIS, BIPOLAR DISORDER, AUTISM SPECTRUM DISORDER, OR TOURETTE SYNDROME

	Log Total Costs			1 (Visited Facility)			1 (Any Harm Behavior)		
	3 Months (1)	1 Year (2)	2 Years (3)	3 Months (4)	1 Year (5)	2 Years (6)	3 Months (7)	1 Year (8)	2 Years (9)
A. Effect of Treatment									
Post-Lasso 2SLS, trimmed sample	-.506	-.087	.239	.490***	.539***	.687***	-.060	-.055	-.024
Table 4 post-Lasso 2SLS estimate	(.329)	(.289)	(.247)	(.067)	(.076)	(.100)	(.060)	(.064)	(.074)
	.884***	.952***	1.085***	.354***	.468***	.631***	.213***	.253***	.273***
	(.303)	(.277)	(.245)	(.078)	(.091)	(.121)	(.055)	(.065)	(.073)
B. Effect of Drug Treatment									
Post-Lasso 2SLS, trimmed sample	-.486**	-.505**	-.482**	.024	-.039	-.101	-.046	-.006	-.083
Table 4 post-Lasso 2SLS estimate	(.220)	(.219)	(.232)	(.053)	(.069)	(.089)	(.037)	(.044)	(.052)
	-.528**	-.501**	-.487**	.021	-.041	-.115	-.042	-.005	-.096*
	(.218)	(.217)	(.233)	(.058)	(.074)	(.093)	(.040)	(.048)	(.057)
C. Effect of Red-Flag Drug Treatment									
Post-Lasso 2SLS, trimmed sample	1.215***	1.374***	1.378***	.236***	.286***	.442***	.147**	.152*	.102
Table 4 post-Lasso 2SLS estimate	(.315)	(.281)	(.273)	(.083)	(.103)	(.119)	(.062)	(.070)	(.079)
	1.135***	1.388***	1.458***	.217***	.266***	.437***	.186***	.180**	.149*
	(.310)	(.286)	(.286)	(.084)	(.103)	(.116)	(.067)	(.074)	(.080)

D. Effect of Gray Area Drug Treatment									
Post-Lasso 2SLS, trimmed sample	-.951***	-1.183***	-1.188***	-.215***	-.260**	-.486***	-.097	-.090	-.055
Table 4 post-Lasso 2SLS estimate	(.308)	(.285)	(.276)	(.083)	(.102)	(.119)	(.066)	(.075)	(.084)
	-.902***	-1.240***	-1.271***	-.179**	-.249***	-.450***	-.089	-.097	-.061
	(.301)	(.295)	(.281)	(.081)	(.101)	(.113)	(.067)	(.077)	(.083)
E. Effect of FDA-Approved Drug Treatment									
Post-Lasso 2SLS, trimmed sample	1.081*	1.255*	1.733**	-.177	-.258	.015	-.225	-.344*	-.310
Table 4 post-Lasso 2SLS estimate	(.628)	(.679)	(.758)	(.175)	(.192)	(.210)	(.157)	(.187)	(.205)
	.898	1.007*	1.625**	-.254	-.306	.063	-.299*	-.378**	-.301
	(.576)	(.606)	(.646)	(.178)	(.195)	(.208)	(.169)	(.190)	(.210)

NOTE.—The table presents results from post-Lasso 2SLS models of mental health treatment on log total health costs (cols. 1–3), an indicator for facility use (cols. 4–6), and an indicator for any harm behavior (cols. 7–9). The trimmed sample estimates exclude children with schizophrenia spectrum and other psychotic disorders, bipolar and related disorders, autism spectrum disorder, and Tourette syndrome from the sample, whereas the table 4 estimates include the full sample. The instruments in the post-Lasso 2SLS models are chosen using a refined data-driven penalty described in Belloni et al. (2012) and are displayed in table A6. Number of instruments for trimmed sample models = 5 (panel A), 3 (panel B), 5 (panel C), 5 (panel D), 1 (panel E). All models include patient controls (age, female, severity, and diagnosis) as well as area, month, and year fixed effects corresponding to the child's initial mental illness claim. Standard errors (in parentheses) are clustered at the three-digit zip code level.

** $p < .05$.
*** $p < .01$.

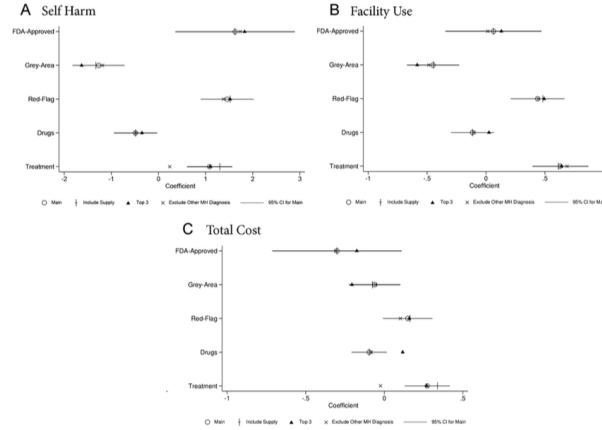


FIG. 6.—Summary of estimated effects of treatment on self-harm, facility use, and total costs. The figure presents results from the post-Lasso 2SLS models of mental health treatment on any self-harm behavior (A), facility use (B), and log total health care costs (C). Circles (“Main”) represent the estimates from our preferred model (as shown in table 4), with the full sample, 116-variable potential instrument vector, and Lasso procedure using the refined data-driven penalty described in Belloni et al. (2012). Lines (“95% CI for Main”) represent the 95% confidence intervals of the “Main” effect coefficient. Triangles (“Top 3”) represent the estimates from an alternative model (as shown in table A4), where Lasso is constrained to select only three instruments. Crosses (“Include Supply”) represent the estimates from an alternative model (as shown in table A5), where a larger potential instrument set—that is, provider supply-side variables (and their interactions with patient characteristics)—is used in the Lasso procedure. Times symbols (“Exclude Other MH Diagnosis”) represent the estimates from an alternative model (as shown in table 6), where children with schizophrenia spectrum and other psychotic disorders, bipolar and related disorders, autism spectrum disorder, and Tourette syndrome are excluded from the sample.

5.7 Table 6 and Figure 6: robustness

Read:

- Table 6 excludes children whose red-flag drugs could plausibly be justified by additional diagnoses such as psychosis, bipolar disorder, autism spectrum disorder, or Tourette syndrome.
- The core patterns survive. Red-flag remains harmful, gray-area continues to lower costs and facility use, and FDA-approved drugs continue to reduce self-harm in the short and medium term.
- Figure 6 shows the main treatment effects are stable across multiple alternative specifications: the preferred instrument set, top-3 instruments only, reintroducing supply variables, and the trimmed sample.
- So the ranking of treatments is not being driven by one fragile first-stage choice.

6 Short summary

- **Method.** The paper studies initial adolescent mental-health treatment using claims data and identifies treatment effects with area-level provider-supply and practice-style instruments selected via post-Lasso 2SLS.
- **Identification.** The key idea is that children in different local treatment markets face different prescribing cultures. These place-based differences move treatment choices even after controlling for diagnosis, severity, demographics, and area/time fixed effects.
- **Application.** The paper compares no treatment, therapy, drug treatment, and within drugs, FDA-approved, gray-area, and red-flag prescribing.
- **Main result.** Red-flag prescribing is clearly associated with worse outcomes. FDA-approved drugs reduce self-harm but are costly. Gray-area prescribing reduces costs and facility use without increasing self-harm.
- **Takeaway.** Professional guidelines and FDA guidance both add value relative to unconstrained discretion, but they do so along different margins.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that treatment guidelines for adolescent mental health are not just symbolic. They have real consequences for children's outcomes.
- In particular, prescribing outside both FDA approval and professional guidelines—red-flag prescribing—appears harmful.
- At the same time, the paper shows that there is a meaningful distinction between strict FDA-based treatment and broader professional-guideline treatment.

7.2 Economic intuition

- FDA approval is based on a narrow but strong evidence base and thus seems especially good at limiting harmful side effects like short-run self-harm.
- Professional societies appear to incorporate a broader clinical perspective, which may make their guidelines more effective at reducing common outcomes such as facility use and overall costs.
- Complete discretion may lead clinicians to choose drugs with weak evidence bases and more dangerous side-effect profiles, which is why red-flag treatment performs poorly.

7.3 Takeaway

This paper is important because it connects a medical treatment problem to a broader economics question about *rules versus discretion*. In this application, some discretion is useful—the gray area supported by professional guidelines appears valuable. But discretion without discipline is costly. The paper's bottom line is that evidence-based treatment guidelines do scale to the real world, and

that large administrative claims data can be used to detect when practice drifts away from good medicine.